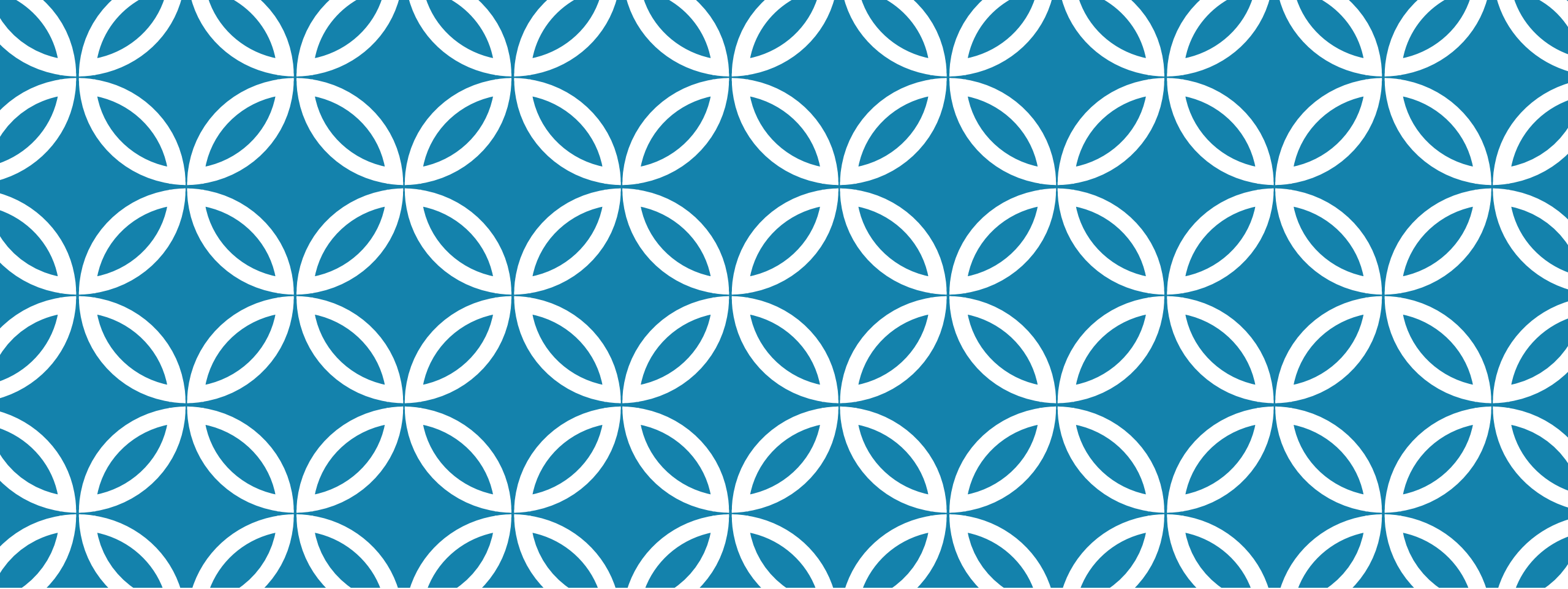


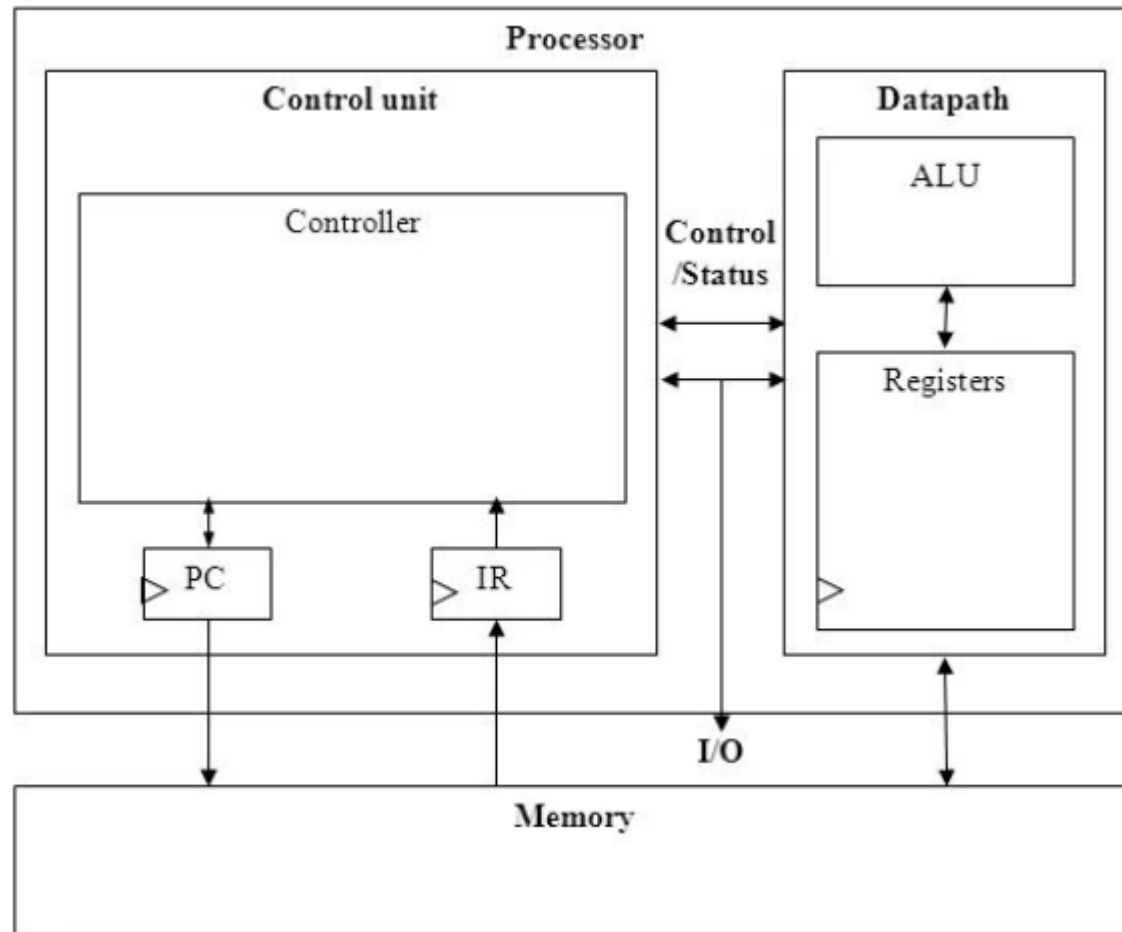


CSG553 RECONFIGURABLE COMPUTING

Networked Embedded Systems & Robotic Applications Group
(NETRA)

Dr. Meetha V Shenoy
BITS Pilani, Pilani Campus
Lect-2

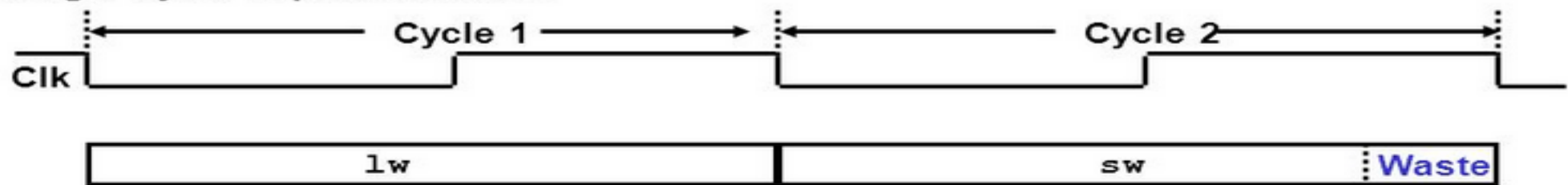
CPU INTERNAL ARCHITECTURE



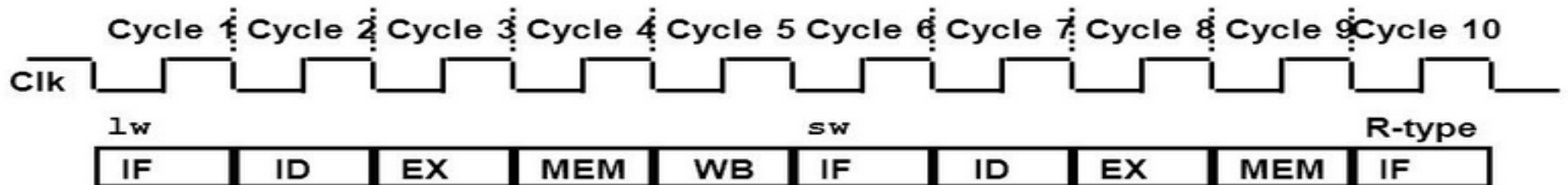
Stored program and data model

SINGLE CYCLE VS MULTICYCLE VS PIPELINED

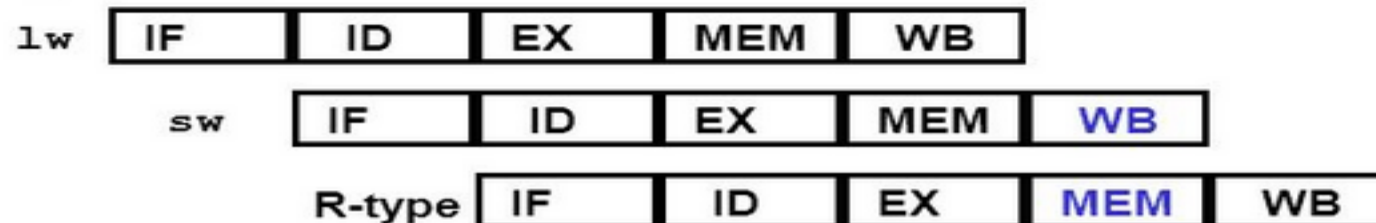
Single Cycle Implementation:

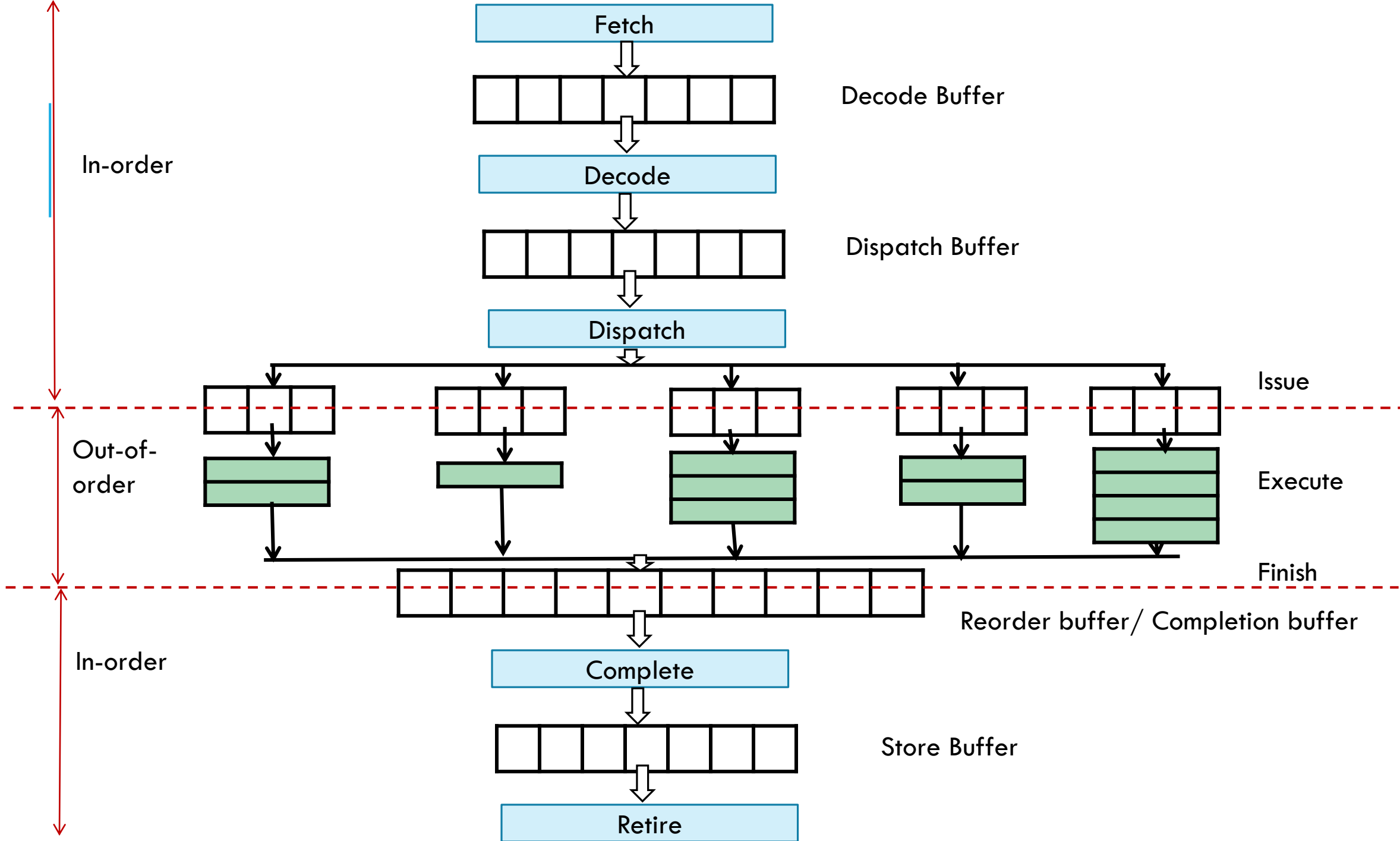


Multiple Cycle Implementation:



Pipeline Implementation:





COMPUTING PARADIGMS

The Von Neumann Computer

General Purpose Processors

Domain Specific Processors/ Application Specific Instruction Set
Processors (ASIP)

Reconfigurable Processors

AUTOMOTIVE NIGHT VISION SYSTEM



COMPUTING PARADIGMS

The Von Neumann Computer

Domain Specific Processors/ Application Specific Instruction Set
Processors (ASIP)

Application Specific Integrated Circuits (ASICs)

Reconfigurable Processors

COMPUTING PARADIGMS

The Von Neumann Computer – Temporal Computation
Reconfigurable Processors

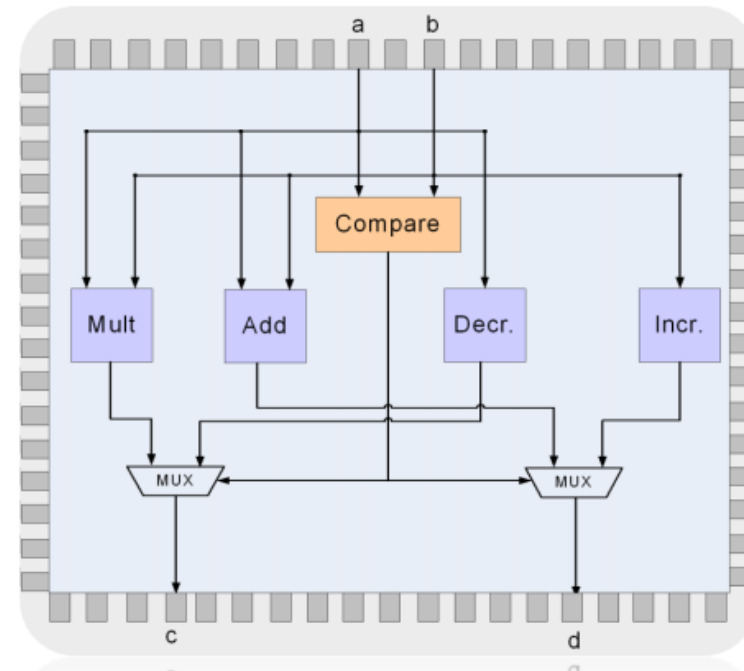
COMPARISON

□ Implementation of a VN computer

```
if (a < b) then
{
    d = a+b;
    c = a*b;
}
else
{
    d = a+1;
    c = b-1;
}
```

Ability to extract parallelism (or concurrency) from algorithm descriptions is the key to **acceleration using reconfigurable computing**

ASIC implementation:

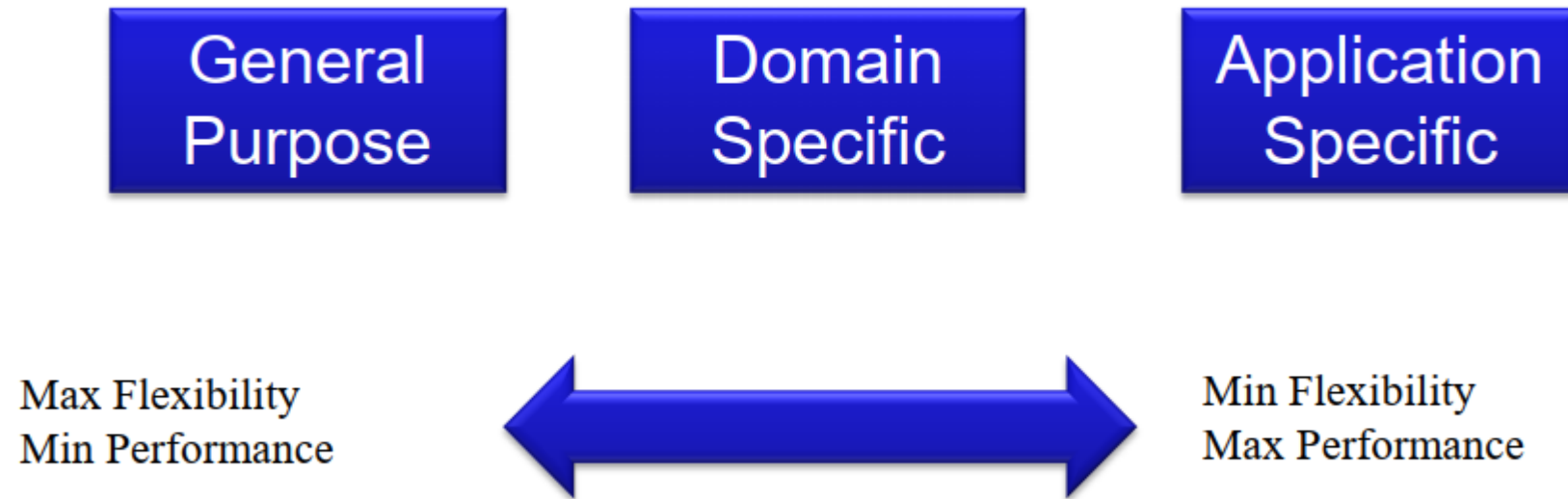


RECONFIGURABLE COMPUTING UNIT

A **Reconfigurable Computing Unit (RCU)** is a specialized hardware component that allows **its internal architecture to be dynamically modified or reprogrammed** after manufacturing, enabling it to perform a wide range of computational tasks efficiently.

- FPGA (Field Programmable Gate Array),
- Coarse-Grained Reconfigurable Arrays (CGRAs),
- Adaptive Compute Acceleration Platform (ACAP)
- Runtime-Reconfigurable Processors, etc.

COMPARISON



RECONFIGURABLE COMPUTING

The Ideal device should combine:

- the flexibility of the Von Neumann computer
- the efficiency of ASICs

SOME APPLICATIONS ???

- ☐ Rapid prototyping
- ☐ Post fabrication customization
- ☐ Multi-modal computing tasks
- ☐ Adaptive computing systems
- ☐ High-performance parallel computing

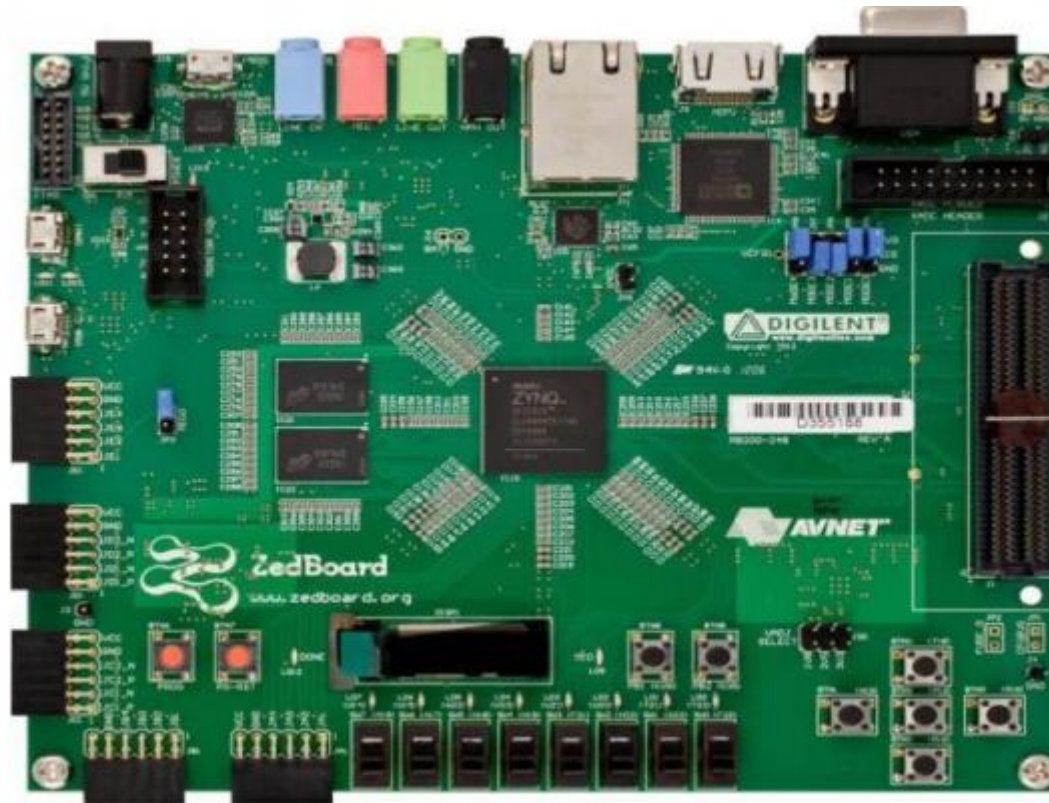
EVALUATION SCHEME

EC No.	Evaluation Component	Duration (min)	Marks (Weightage %)	Date and Time	Nature of Component
1	Mid-Semester Exam	90 Min	25%		Closed Book
2	Regular Labs/ Lab Assignments	NA	20%	Regular/ To be announced	Open Book
3	Project	NA	20%	Regular/ To be announced	Open Book
4	Comprehensive Exam	3hrs	35%		Closed Book

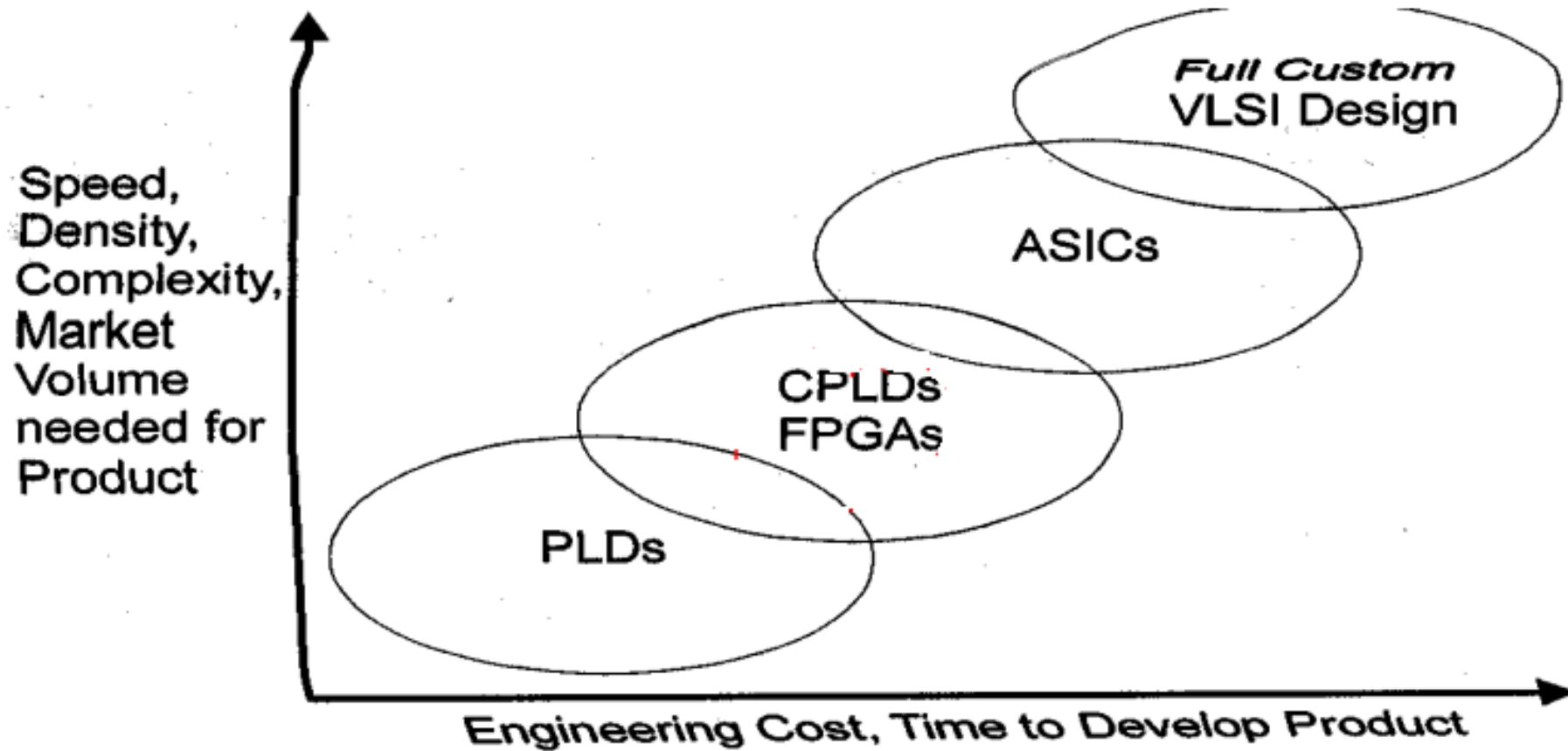
LAB

Understanding of Verilog mandatory

Xilinx Vivado, Zedboard



DIGITAL LOGIC



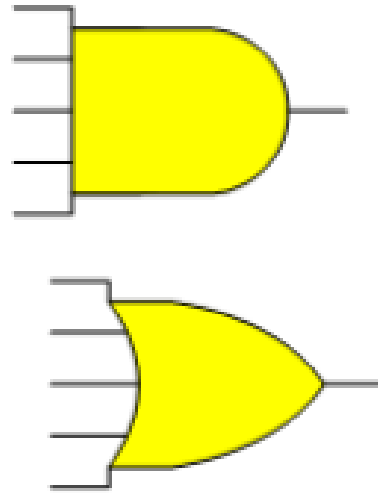
PROGRAMMABLE LOGIC DEVICES (PLD)

PROM

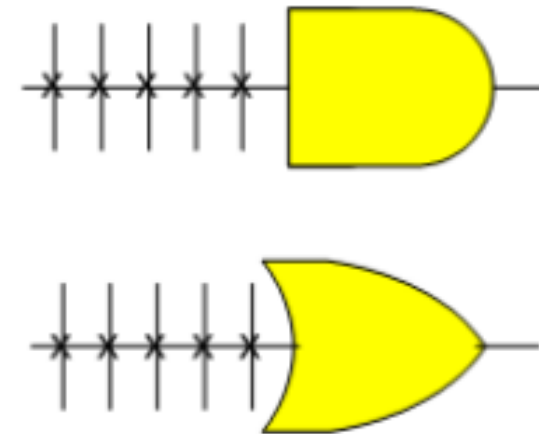
Programmable Logic Array (PLA)

Programmable Array Logic (PAL)

ARRAY LOGIC REPRESENTATION

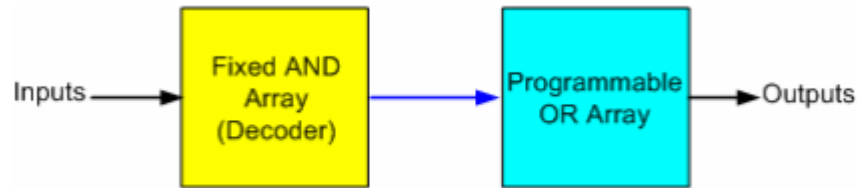


(a) Conventional
Symbol

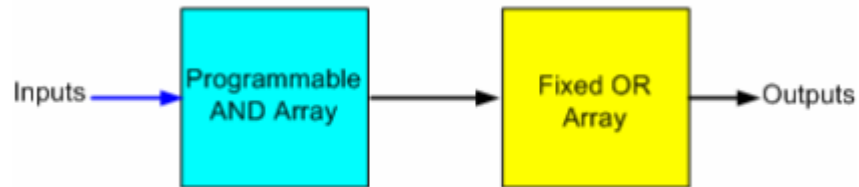


(b) Array Logic
Symbol

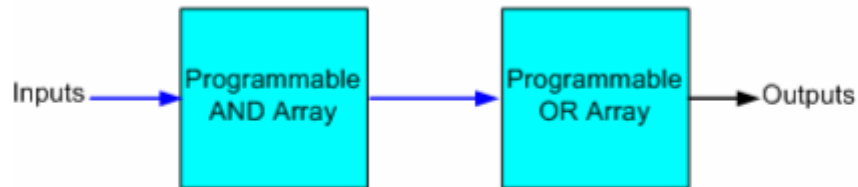
FUNDAMENTAL PLD TYPES



(a) Programmable Read Only Memory (PROM)



(b) Programmable Array Logic (PAL) Device

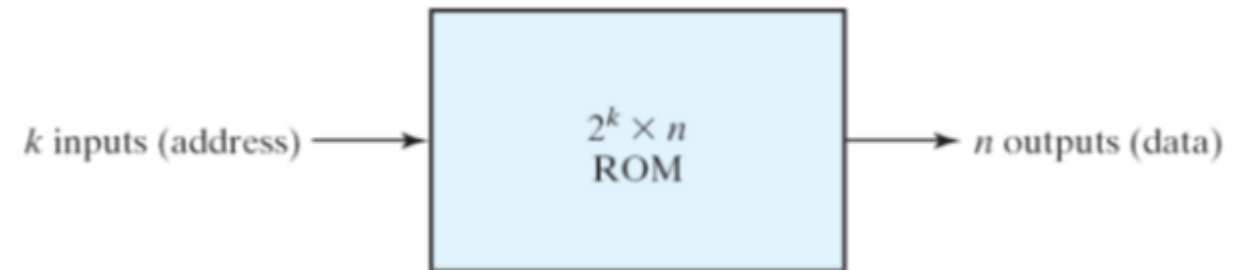


(c) Programmable Logic Array (PLA) Device

→ Programmable Connections

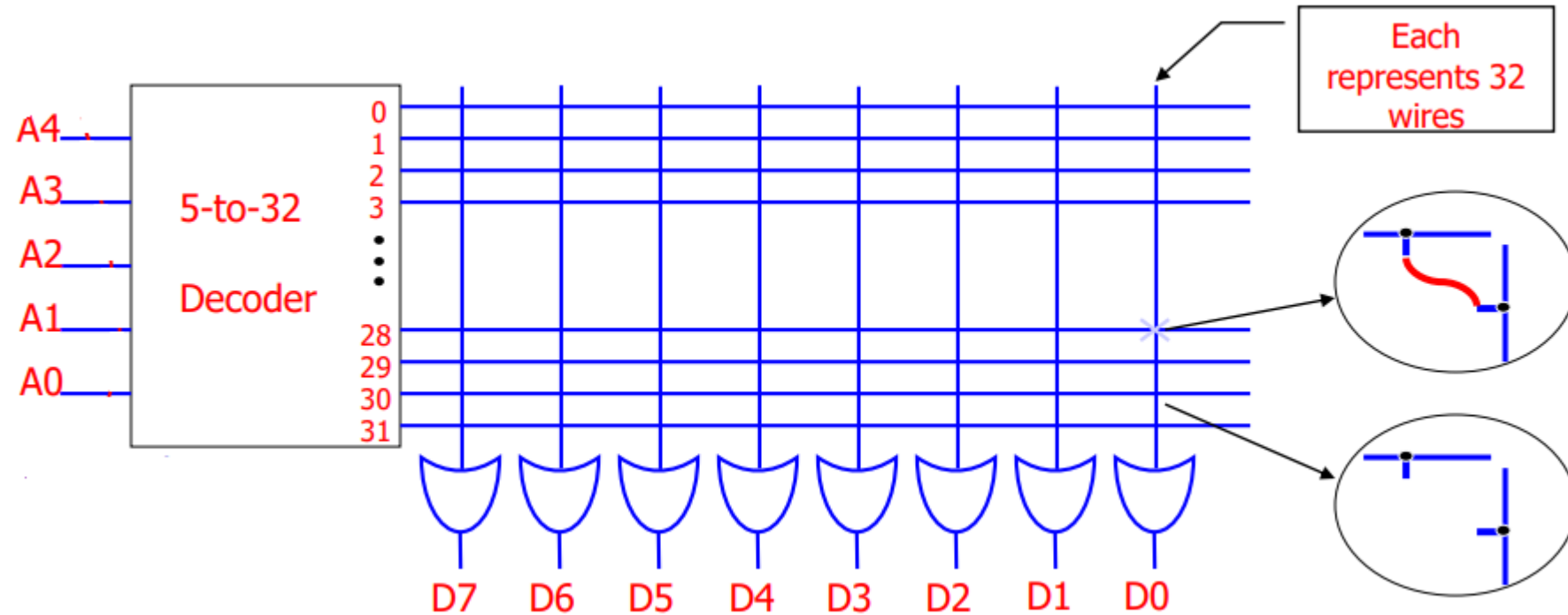
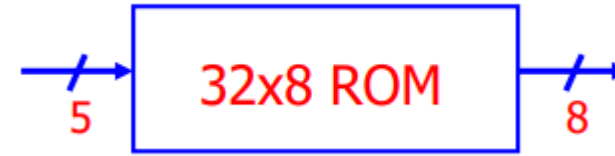
→ Normal Connections

PROM



No data inputs, there may be enable inputs

32X8 ROM



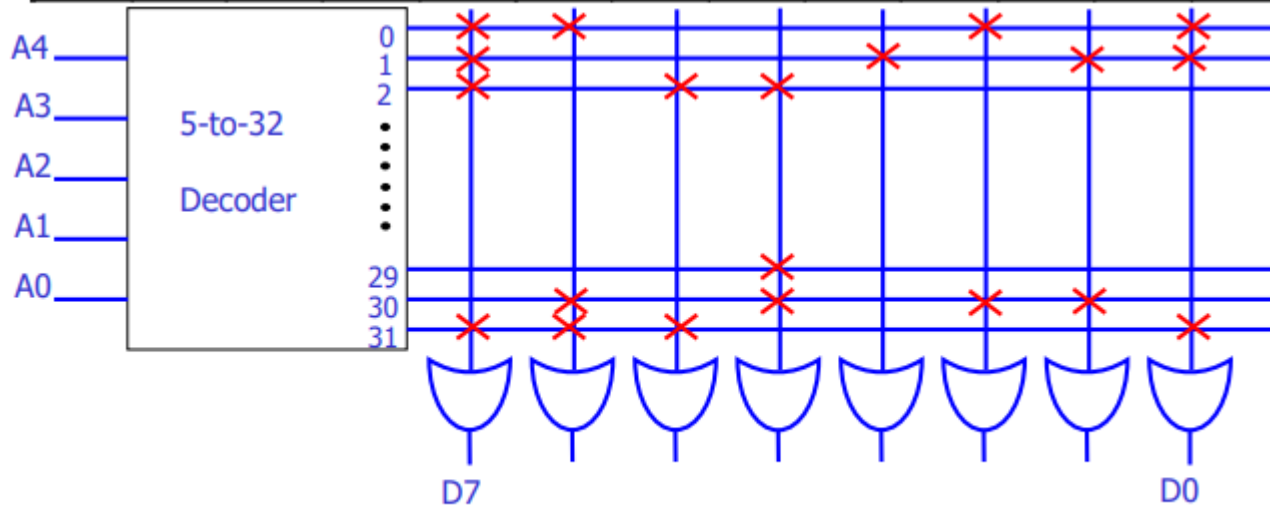
DECODER

3 to 8 Line Decoder

Inputs			Outputs							
x	y	z	D0	D1	D2	D3	D4	D5	D6	D7
0	0	0	1	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0	0	0	0
0	1	0	0	0	1	0	0	0	0	0
0	1	1	0	0	0	1	0	0	0	0
1	0	0	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	0	1	0	0
1	1	0	0	0	0	0	0	0	1	0
1	1	1	0	0	0	0	0	0	0	1

32X8 ROM

A4	A3	A2	A1	A0	D7	D6	D5	D4	D3	D2	D1	D0
0	0	0	0	0	1	1	0	0	0	1	0	1
0	0	0	0	1	1	0	0	0	1	0	1	1
0	0	0	1	0	1	0	1	1	0	0	0	0
...	...	...	...	...	...	...	...	...	...	...	...	...
1	1	1	0	1	0	0	0	1	0	0	0	0
1	1	1	1	0	0	1	0	1	0	1	1	0
1	1	1	1	1	1	1	1	0	0	0	0	1

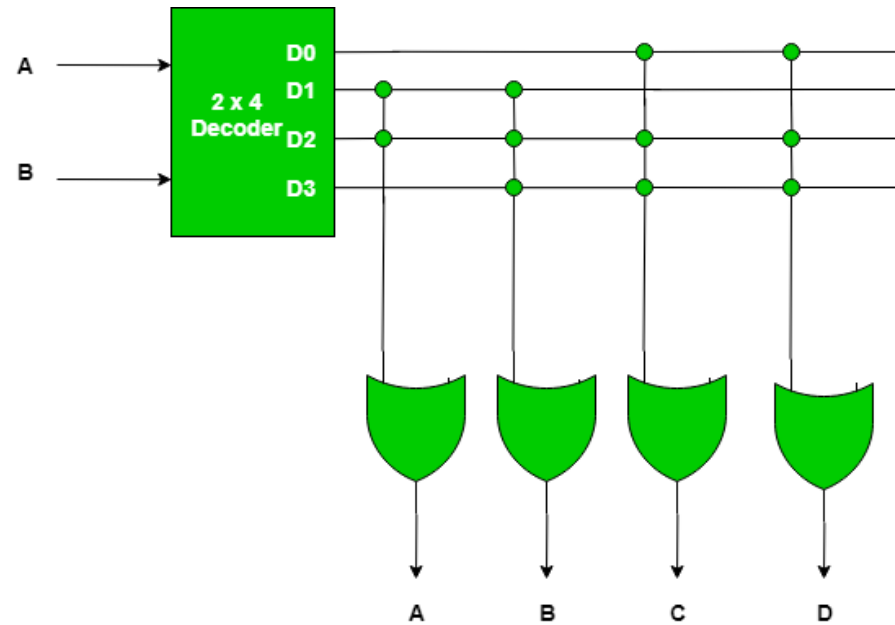


X indicates a connection

4X4 ROM

Size of decoder ?

No of OR gates



Implement 4 functions using
2 inputs

DESIGN A SQUARE LOOKUP TABLE FOR $F(X) = X^2$ USING ROM

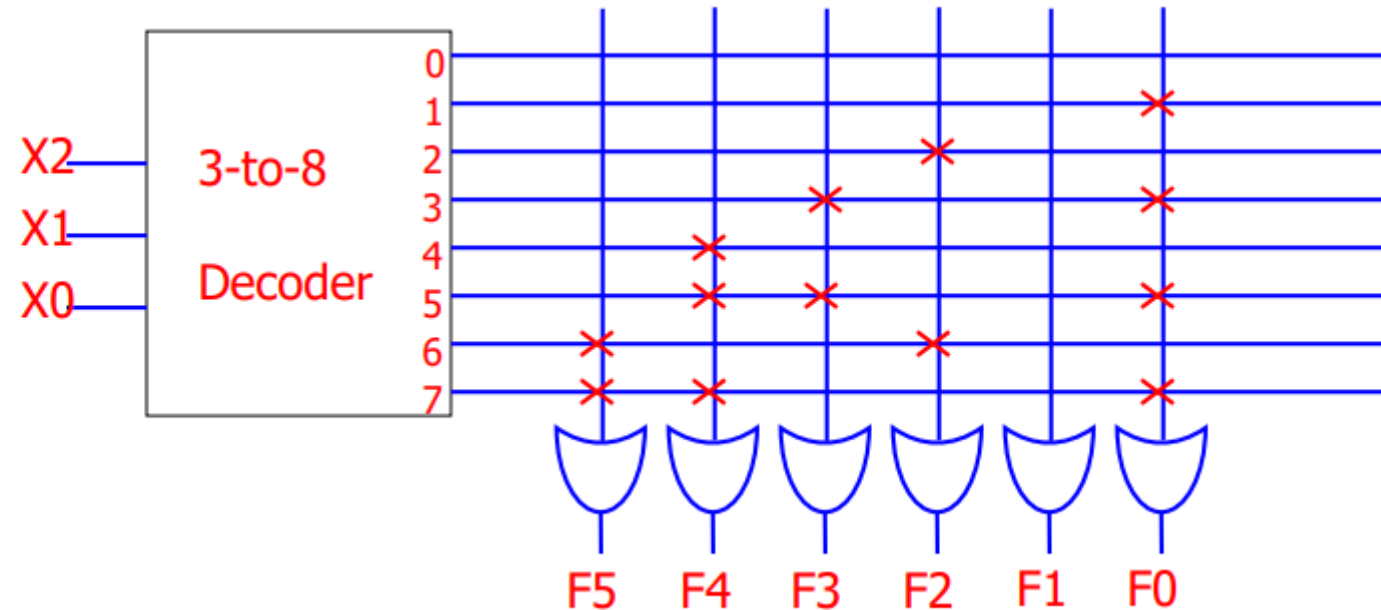
X	$F(X)=X^2$
0	0
1	1
2	4
3	9
4	16
5	25
6	36
7	49



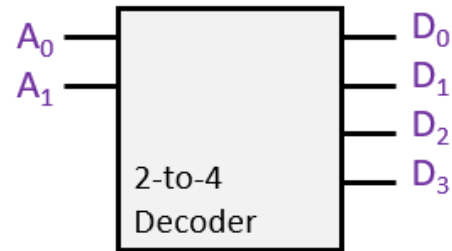
X	$F(X)=X^2$
000	000000
001	000001
010	000100
011	001001
100	010000
101	011001
110	100100
111	110001

DESIGN A SQUARE LOOKUP TABLE FOR $F(X) = X^2$ USING ROM

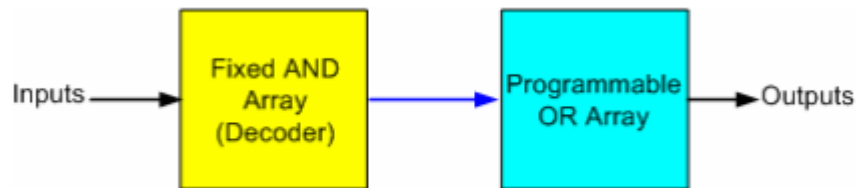
X	$F(X)=X^2$
000	000000
001	000001
010	000100
011	001001
100	010000
101	011001
110	100100
111	110001



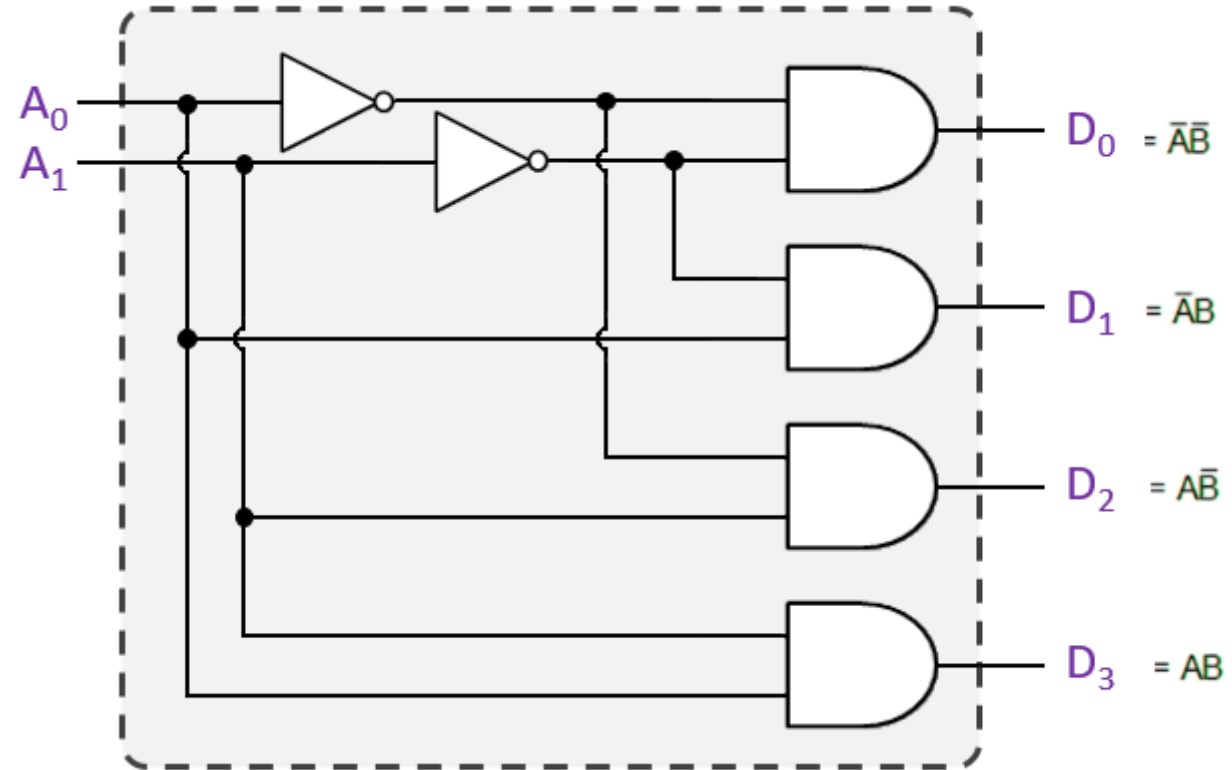
ROM AS A PLD



INPUT A_1A_0	OUTPUT $D_3D_2D_1D_0$
00	0001
01	0010
10	0100
11	1000



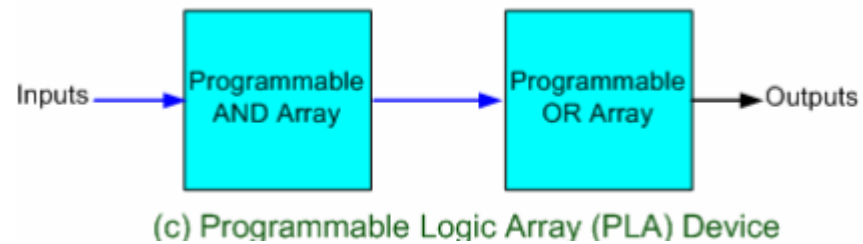
(a) Programmable Read Only Memory (PROM)



→ Programmable Connections

→ Normal Connections

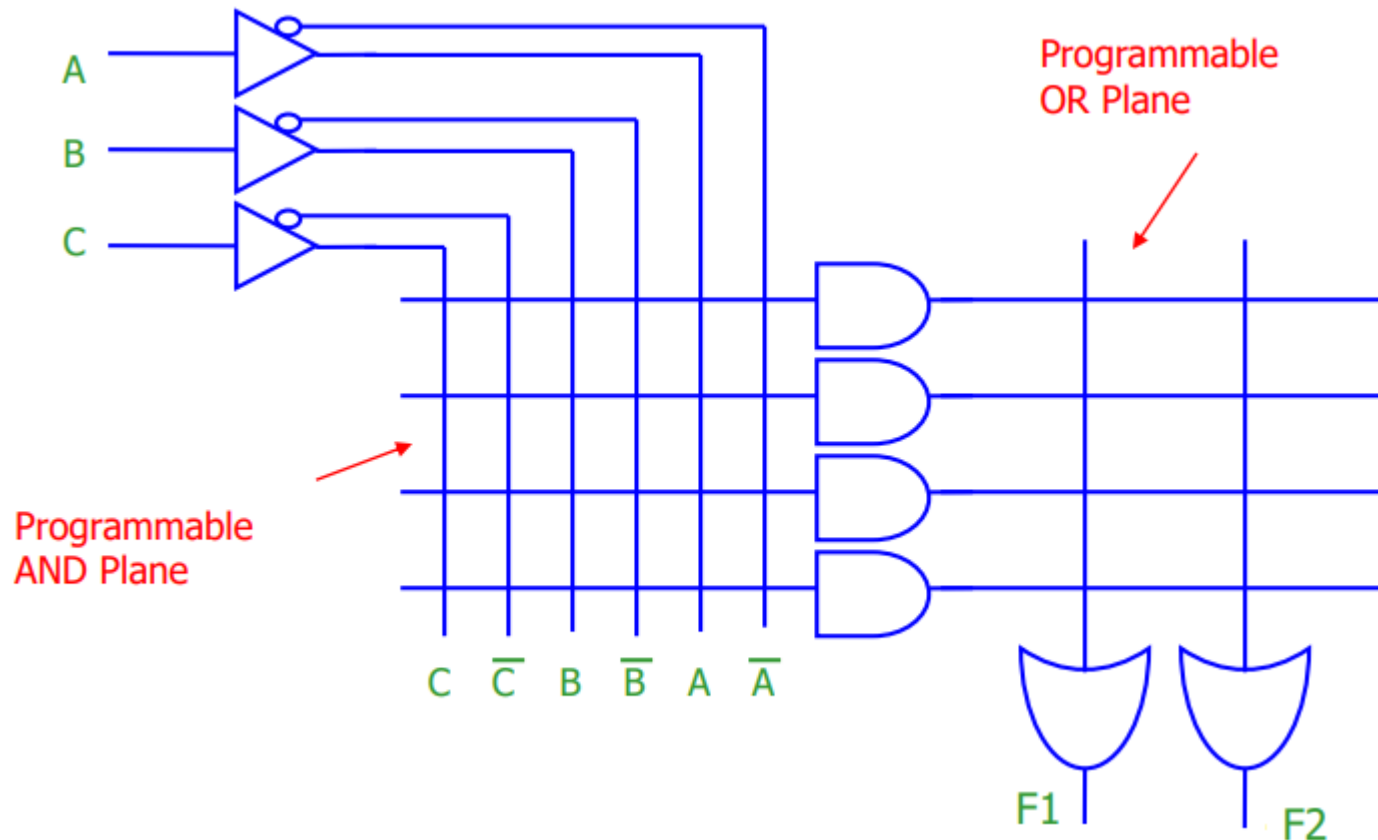
PROGRAMMABLE LOGIC ARRAY



→ Programmable Connections

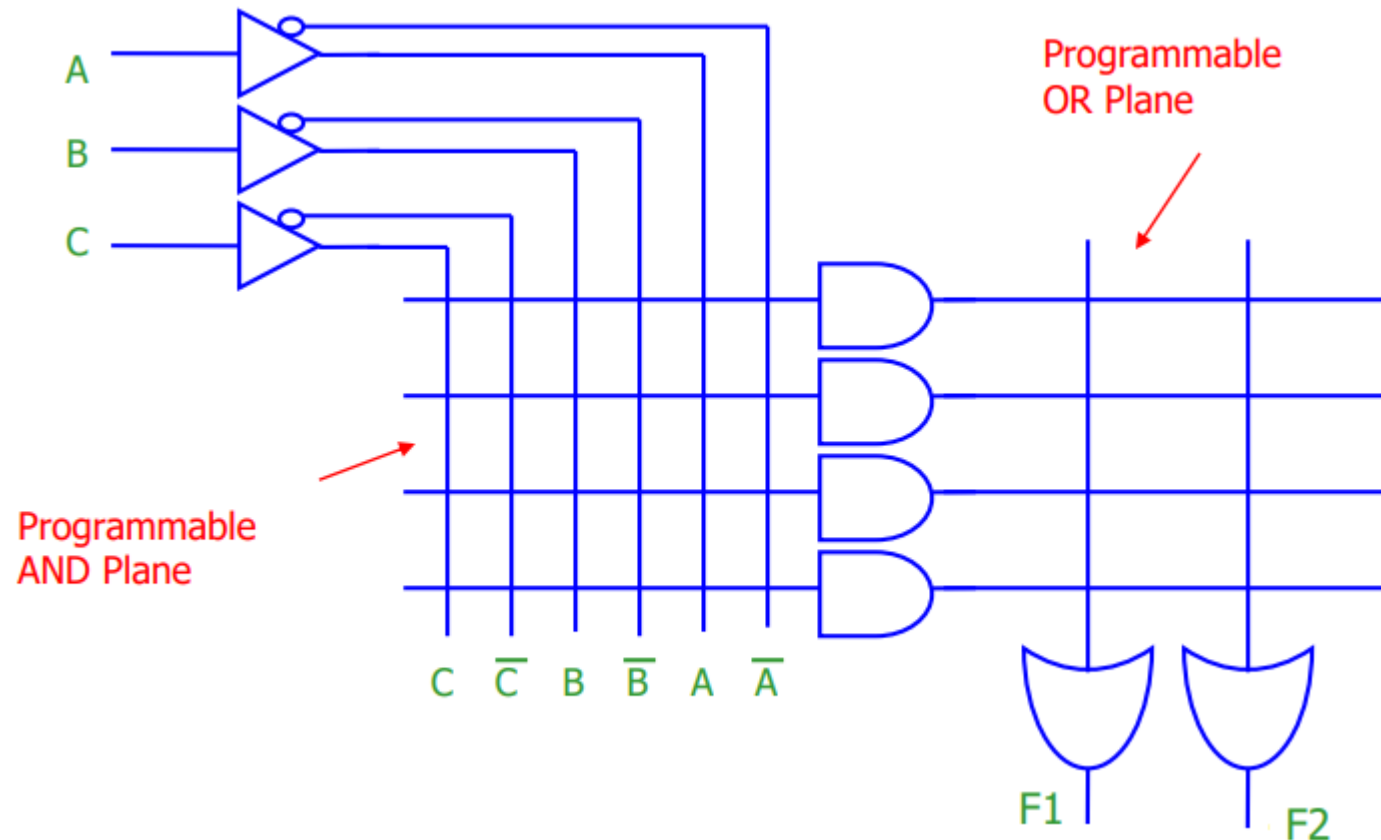
→ Normal Connections

PLA



SIZE OF PLA: 3X4X2
3 INPUTS
4 PRODUCT TERMS
2 OUTPUTS

OBTAIN TRUE AND COMPLEMENTARY OUTPUT



A	B	Q
0	0	0
0	1	1
1	0	1
1	1	0

$$V2 = \Sigma m(1,2,3,4) = xy'z' + x'z + x'yz'$$

$$V1 = \Sigma m(2,6,7) = x'yz' + xy$$

$$V0 = \Sigma m(4,6,7) = xy'z' + xy$$

Size= 3x4x3

